

Visual Regression Testing

Automated visual comparison to detect unexpected layout and rendering changes.

What it is

Visual Regression compares a current UI image with an approved baseline and highlights pixel or region differences.

Where it helps

Design systems, stable pages, reusable components, responsive layouts and multiple browsers can benefit from visual comparisons.

False positives

Fonts, animations, dates, dynamic images, anti-aliasing and variable content can create differences without representing real defects.

Strategy

Use controlled baselines, mask dynamic regions, justify tolerances and review differences before updating the reference.

When to use it

Use this knowledge when product risk, architecture or the delivery stage requires this type of validation. Confirm the goal, environment, required data and expected evidence before starting.

How QA applies it

Start from the behavior or risk to validate, prepare data and environment, execute the technique or tool reproducibly and capture evidence that helps the team understand the result.

QA tip

Never update a baseline just to make a test pass; first confirm that the visual change is expected.